Social Sciences & Humanities Open 12 (2025) 101809

Contents lists available at ScienceDirect

Social Sciences & Humanities Open

journal homepage: www.sciencedirect.com/journal/social-sciences-and-humanities-open

Regular Article

Exploring public sector workforce attitudes toward AI competence and its impact on AI competence self-assessment

Gatis Lāma [a], Agnese Lastovska [b,*]

[a] Faculty of Social Sciences, Rīga Stradiņš University, 1019, Riga, Latvia
[b] University of Latvia, Riga, Latvia

ARTICLE INFO

Keywords:
Artificial Intelligence (AI)
AI competence
Public sector workforce
AI ethics

ABSTRACT

This study addresses the growing importance of AI competence in the public sector, particularly in Latvia, where AI adoption is increasingly relevant for improving workplace efficiency and decision-making. The research explores self-assessed AI competence among public sector employees, focusing on the impact of workplace training and trust in AI-generated information. AI competence was evaluated using 10 sub-competencies based on AI functionalities, assessed on a 7-point Likert scale. A total of 1557 participants contributed to the study, which was analysed using descriptive statistics, Cronbach's alpha, Spearman rank correlation, network analysis, and the Kruskal-Wallis H test. The results show that only three AI sub-competencies scored close to the scale's mean value, indicating the limited use of advanced AI functions. Employees who received AI training rated their overall competence significantly higher than those without training. Additionally, more than half of the respondents expressed limited trust in AI-generated information, with only seven participants fully trusting AI. These findings underscore the need for targeted AI training to enhance competence and trust, highlighting the importance of integrating AI tools effectively in the public sector. The data that support the findings of this study are available on request from the corresponding author.

1. Introduction

The development of Europe's artificial intelligence (AI) strategy encompasses a wide range of dimensions – such as ecosystems, education, governance, liability, data access, and digital autonomy. These dimensions are critical for aligning stakeholder needs and perceptions, which are essential to the successful integration of AI across the European landscape (Polyviou & Zamani, 2023). AI is rapidly transforming business processes by enhancing efficiency, improving decision-making and solving complex problems (Bormane & Blaus, 2024; Nag et al., 2023). The anticipated boost in global GDP by 2030 due to AI adoption is compelling businesses to rethink models and embed AI strategies at their core (Echeberria, 2022; Livanis et al., 2023).

Beyond the private sector, AI is also being applied in public administration to improve service efficiency, although implementation remains challenging due to ethical concerns and workforce readiness (Labanava et al., 2022). In the Baltic states, including Latvia, progress is being made in integrating AI into the judiciary through regulatory reforms and digital tools (Limante & Sukyte, 2025). Healthcare, too, is embracing AI technologies, with promising applications in medical diagnostics such as fracture detection (Boginskis et al., 2023). Meanwhile, initiatives such as the RUTA:MED platform demonstrate how domain-specific language resources can enhance the accuracy of medical speech recognition (Auziņa et al., 2022).

AI is also influencing public education. At the Latvia University of Life Sciences and Technologies, AI tools are being incorporated into English for Specific Purposes (ESP) courses, with students reporting benefits for learning and digital skill development (Sinkus & Ozola, 2024). In the field of entrepreneurship education, AI is helping to unlock the entrepreneurial potential of Latvian students (Voronov et al., 2023). Similarly, robotic process automation is being explored to optimise e-invoicing systems as part of the country's shift toward a digital economy (Gunaratne & Pappel, 2020).

AI's impact extends to migration management and border security, where it is used to address challenges arising from geopolitical crises, such as the war in Ukraine (Szwed, 2022). Latvia is aligning its national policies with EU regulations such as the AI Act, which emphasise secure, ethical and human-centred AI deployment (Bostan, 2025; Ersoy, 2024; Olimid et al., 2024).

Public administration in Latvia is beginning to experiment with AI

* Corresponding author.
E-mail addresses: gatis.lama@rsu.lv (G. Lāma), agnese.lastovska@lu.lv (A. Lastovska).

https://doi.org/10.1016/j.ssaho.2025.101809
Received 4 November 2024; Received in revised form 16 July 2025; Accepted 16 July 2025
Available online 19 July 2025
2590-2911/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

tools such as e-government chatbots and analytics systems (United Nations, 2022). Yet, successful adoption depends on organisational culture and workforce readiness. Research highlights that innovation-supportive cultures are more likely to foster competence development, while rigid structures may suppress it (Höchtl et al., 2016).

While the term "artificial intelligence" continues to evolve, major institutions offer working definitions. The OECD defines AI as machine-based systems that make predictions or decisions to affect real or virtual environments (Berryhill et al., 2019). The European Commission views AI as autonomous systems that pursue goals based on environmental input (European Commission, 2019). These evolving definitions underscore AI's complexity and its wide-ranging implications for different sectors of society.

Despite AI's longstanding presence in research, public sector adoption has historically lagged behind that of the private sector (Wirtz et al., 2018). Nevertheless, growing policy momentum is visible: over 50 countries have launched or planned national AI strategies, with 36 including public-sector priorities (Berryhill et al., 2019; OECD, 2024a, 2024b). As the global workforce adapts to rapid digitalisation and automation, the demand for reskilling has intensified (Eurofound, 2021; Schwab & Zahidi, 2020). Private enterprises have generally moved more quickly to address these needs, investing in AI-focused training and corporate academies.

In contrast, many public institutions face challenges in identifying relevant AI skills and providing structured learning opportunities. Recent studies show that while private sector employees feel increasingly prepared for AI-related roles, public employees often lack the same support systems (McKinsey & Company, 2025; Chinn et al., 2020). Organisational digitisation plays a key role: institutions with more advanced digital infrastructure are significantly more likely to have embedded AI into core processes.

The public sector must therefore navigate not only technological gaps but also ethical, privacy and transparency concerns – especially in sensitive areas like healthcare and justice (OECD, 2024a, 2024b). International efforts such as UNESCO's Digital Transformation and AI Competency Framework for Civil Servants aim to address these barriers by emphasising trust, inclusivity and sustainability (Calvino & Fontanelli, 2023).

Given the limited research on AI competence within the public sector, this study aims to examine the self-assessed AI competence of public sector employees in Latvia. It specifically investigates the extent to which workplace support influences perceived competence levels, and how this competence relates to employees' trust in AI-generated information as well as their awareness of ethical issues associated with the use of AI tools in professional settings.

1.1. Literature review

1.1.1. Understanding competence in the context of AI

The concept of competence has long been studied in both education and workforce development, with historical roots tracing back thousands of years. Yuanjing Wilcox (2012) identified early notions of competence in ancient Chinese society, while Käthe Schneider (2019) emphasised its presence in the legal frameworks of Ancient Rome. These historical perspectives illustrate that competence has been central to human development and societal functioning, particularly within the realms of governance and law. In modern contexts, competence is generally understood as a multifaceted construct that integrates knowledge, skills, attitudes, and motivations. A widely recognised definition from Cambridge Assessment describes competence as "the ability to integrate and apply contextually-appropriate knowledge, skills and psychosocial factors (e.g., beliefs, attitudes, values, and motivations) to consistently perform successfully within a specified domain" (Vitello, Greatorex, & Shaw, 2021). This framing highlights that competence is not only about what one knows or can do but also about how one approaches tasks – confidently, ethically and with motivation.

In organisational contexts, the development of competence is closely tied to performance and productivity (Slišāne et al., 2021). McClelland (1993) argued that competence is a key driver of effective job performance, a view reinforced by Salman et al. (2020), who emphasised that structured human resource strategies – such as on-the-job training and continuing education – can significantly enhance both individual and organisational outcomes. Competence thus acts as a strategic lever in aligning employee capabilities with organisational goals (Slišāne et al., 2022). Van der Klink and Boon (2002) similarly point out that well-defined competencies help bridge the gap between educational outcomes and labour market demands. Hartle (1995) adds that in addition to knowledge and skills, character traits and personal motivation are crucial for competence to translate into workplace success.

In the context of Artificial Intelligence (AI), competence requires a nuanced blend of cognitive, technical and ethical dimensions. Competence in AI involves knowing what AI tools exist and how they function, being able to apply them effectively in work-related tasks, and maintaining an attitude of curiosity, responsibility and critical engagement. As noted by recent frameworks on AI literacy and digital competence, traditional digital skills models are only beginning to incorporate AI-specific components (Chiu et al., 2024). UNESCO's AI competency framework for educators, for example, underscores the importance of not only technical skills but also a human-centred mindset, ethical reasoning and responsible use (UNESCO, 2021).

To systematically assess AI competence, this study draws on Bloom's taxonomy – a widely used framework that classifies cognitive objectives from basic to advanced: remember, understand, apply, analyse, evaluate, and create. In our study, Bloom's levels were adapted to measure the depth of engagement with AI tools through a seven-point scale:

0 – "I haven't even heard of the relevant AI tools"
1 – "I have heard of the tools but haven't used them"
2 – "I know how to use the tools, but haven't used them in practice"
3 – "I have used the tools for simple tasks"
4 – "I analyse my work to deliberately select appropriate AI tools for tasks"
5 – "I can evaluate and combine different AI tools for my needs"
6 – "I improve or customise AI tools, or integrate them into other systems I use"

This structured, developmental scale not only assesses whether employees have used AI tools but also reveals the sophistication and intentionality behind their usage. Higher-order competencies – such as adapting and creating with AI – are qualitatively distinct from basic awareness and are typically only achieved with sustained training and practical exposure. In nascent domains like AI, especially within the public sector, such higher-level competence is often rare due to limited training opportunities (Wirtz et al., 2018). Using Bloom's taxonomy ensures that competence is measured not just as binary (yes/no) usage but as a gradient of mastery (Anderson et al., 2001). This approach enables organisations to identify specific skill gaps: for example, while many employees may be aware of AI-based translation services, far fewer are integrating multiple AI tools to automate complex workflows. Simpler tools like translation and classification systems often serve as entry points due to their accessibility (Wirtz et al., 2018), while more advanced applications – e.g., predictive analytics – require greater skills and investment, leading to slower uptake (Pero et al., 2022).

Hypothesis 1. Competence Variation by AI Function

We hypothesise that public sector employees' AI competence will vary across different functional domains – being relatively higher for common, familiar applications (e.g., information retrieval or translation) and lower for more complex, specialised functions (e.g., predictive analytics or automation).

This expectation aligns with prior research suggesting that AI technologies tend to gain initial traction in applications that support routine

tasks and improve everyday efficiencies. In contrast, advanced applications require more technical expertise and are less likely to be adopted without targeted training. In other words, we anticipate that self-assessed competence will be higher for simpler AI tasks that demand lower skill thresholds and have been available for longer, compared to cutting-edge AI capabilities that are still emerging within the public sector.

1.1.2. The role of artificial intelligence in workforce competence

The integration of artificial intelligence (AI) into public sector organisations can be effectively analysed through technology adoption and innovation theories. One widely recognised framework is the Unified Theory of Acceptance and Use of Technology (UTAUT), which identifies four core determinants of technology adoption: performance expectancy (belief in performance enhancement), effort expectancy (ease of use), social influence (peer and organisational expectations), and facilitating conditions (technical and institutional support) (Nillos, 2016).

In the context of AI, performance expectancy reflects employees' beliefs about how AI can enhance efficiency and decision-making. Effort expectancy pertains to how user-friendly AI tools are perceived. Facilitating conditions include access to infrastructure, training and supportive policies, while social influence may stem from leadership and organisational culture. However, social norms in technology adoption are often shaped by contextual complexities (Venkatesh et al., 2003, 2012).

An essential extension of UTAUT in AI research is the central role of trust. Due to AI's often opaque, "black box" nature, users may hesitate to accept its outputs. Trust – shaped by perceptions of accuracy, transparency and fairness – has emerged as a critical determinant of AI adoption (Glikson & Woolley, 2020). Low trust reduces engagement, limiting opportunities to build competence. Conversely, users with positive AI experiences or greater understanding are more likely to engage and develop competence. Trust, therefore, is not blind acceptance but pragmatic confidence in AI's reliability and ethical alignment (Durán & Pozzi, 2025; Siau & Wang, 2018).

Public sector innovation theory further emphasises the importance of organisational transformation, skill development and process redesign for successful AI implementation (Borins, 2001; De Vries et al., 2016). OECD reports highlight that AI adoption reshapes workforce requirements, making structured skill-building essential (OECD, 2024a, 2024b). In Latvia and elsewhere, governments are piloting AI tools such as e-government chatbots and analytics platforms (United Nations, 2022). These initiatives rely heavily on employee readiness and organisational culture. Cultures that foster learning and innovation facilitate competence development, whereas rigid environments may inhibit it (Höchtl et al., 2016).

At every level, trust is foundational: employees must trust AI tools, managers must trust employees to use them ethically, and the public must trust in the government oversight of AI (Glikson & Woolley, 2020). Training enhances technical capabilities and lowers perceived barriers (Venkatesh et al., 2003), while trust elevates performance expectations and supports adoption.

The rise of AI coincides with global shifts in workforce demands, emphasising adaptability, collaboration and digital fluency (Arrighi, 2005; Grant et al., 1979). While the private sector often leads in AI implementation (Kaplan & Haenlein, 2019), the public sector must also keep pace. Since the 2015 AI resurgence, innovations have transformed sectors such as healthcare and administration (Ting et al., 2019), though employee preparedness remains uneven. Emotional and cognitive responses – ranging from excitement to scepticism – can strongly affect engagement (Waytz et al., 2010).

As AI adoption matures, tailored upskilling and reskilling strategies become vital (Baki et al., 2023). Digital skills training should integrate support for cognitive, emotional and social well-being (Cramarenco et al., 2023). Learning organisations must embed AI responsibly and inclusively into their structures. For example, Australia identifies skill gaps as a persistent barrier to AI progress (Loucks et al., 2019). Moreover, flexible work arrangements enhance satisfaction and performance, especially under digital and hybrid models (Davidescu et al., 2020). Low-skill roles are more susceptible to automation, while high-skill roles require the continual renewal of expertise (Dosi et al., 2021).

Critically, AI is not merely a labour-reducing tool – it creates new opportunities. The concept of "creative destruction" explains this duality: innovation disrupts old roles while generating new ones. Realising this potential demands strategic policies, inclusive structures and lifelong learning systems (Nübler, 2016). Organisations must foster environments that promote responsible, inclusive and strategic AI adoption – combining technical training with human-centred design, collaboration and long-term capacity-building (Alnamrouti et al., 2022).

Ultimately, AI is reshaping both public and private sector realities. The key lies not only in technology acquisition but also in cultivating the mindsets, trust and competencies required to use it effectively. For the public sector, this entails investment in training, trust-building and cultural transformation – all of which are essential for developing next-generation, AI-enabled governance.

H2. AI Training Effect. Public sector employees who have received AI training will report significantly higher AI competence than those without such training.

This hypothesis reflects UTAUT's concept of facilitating conditions and is supported by findings that structured training enhances workplace AI capabilities.

Research supports this: individuals with AI training report stronger self-efficacy and motivation for AI use (Oran, 2023), as well as better pedagogical understanding (Celik, 2023; Nazaretsky et al., 2022; Cavalcanti et al., 2021). Teachers with AI training also show higher behavioural commitment and intrinsic motivation (Bruna et al., 2023; Postareff et al., 2007; 2008).

H3. Trust Effect. Public sector employees with greater trust in AI-generated information will report higher self-assessed AI competence than those with low trust.

Trust in technology correlates strongly with the intention to use and learn new tools. Trust boosts engagement, enabling skill development.

Individual differences – such as personality and self-efficacy – influence trust (Becker & Fischer, 2024). Competence is a critical component of trust: high AI competence enhances user trust and adoption (Gonzalez et al., 2025; Li et al., 2025), while perceptions of AI expertise, humanness and congruence affect user acceptance (Krop et al., 2024). Integrity and fairness also shape trust (Mehrotra et al., 2024), and perceived competence plays a key role in trust calibration – the process of aligning user trust with system performance (Shafi, 2017).

Explainable AI improves transparency, thereby boosting trust and perceived competence (Eke & Shuib, 2024). Ultimately, competence and trust are mutually reinforcing in successful AI adoption.

2. Materials and methods

The research data were collected using the survey tool QuestionPro. The survey consisted of demographics, geography, socio-cultural questions, AI sub-competence self-assessment statements, and questions about attitudes towards AI and educational opportunities in the workplace. AI competence was defined by 10 AI sub-competencies based on AI functionality, with each functionality measured by one statement on a 7-point Likert scale, adapted from Bloom's taxonomy (0 – I haven't heard about the appropriate AI tools, 1 – I have heard something about the appropriate AI tools but haven't used them, 2 – I have heard about the appropriate AI tools and know how to use them, 3 – I have used the appropriate AI tools for simple tasks, 4 – I deliberately analyse my daily work and select the most appropriate AI tools, 5 – I evaluate and combine different AI tools, 6 – I improve the appropriate AI tools or

integrate them into other systems).

Attitudes towards AI and educational opportunities were measured using statements on a 4-point Likert scale (4 – Yes, 3 – Rather yes, 2 – Rather no, 1 – No). The survey was distributed via email. Public sector employees' email addresses were gathered from official municipal websites, and an email with an invitation to participate, along with a link to the online survey, was sent to all employees' emails. In total, 11,302 emails were sent, and 1557 public sector employees participated in the study (Table 1).

Non-experimental research design was chosen to describe existing phenomena without manipulating conditions to affect subjects' responses (Govindan, 2013). Cronbach's alpha values were calculated to determine the reliability of the Likert scales. Data were analysed using descriptive statistics. The Kruskal-Wallis H test was performed to determine whether there are statistically significant differences in AI competence based on different attitudes towards AI, and whether there are statistically significant differences between AI competence and education opportunities in the workplace to test the study's hypothesis. This analytical approach is appropriate for non-parametric data (Ostertagová et al., 2014). Additionally, a Spearman rank correlation test was conducted to determine whether there is a correlation between AI sub-competencies and network analysis was used with the glasso (or graphical lasso) procedure to determine a network with edges defined by partial correlation coefficients to further reveal interrelations (Leme et al., 2020) of AI competence and the importance of AI training. The questionnaire was available for completion from July 6, 2024 to August 30, 2024, and the data were analysed using SPSS version 29 and Microsoft Excel. The study adhered to all ethical research standards in accordance with the General Data Protection Regulation (GDPR). Participants completed the questionnaire anonymously, and participation was entirely voluntary. Approval for conducting this research was obtained on June 12, 2024 from the Research Ethics Committee of Social Sciences and Humanities of the University of Latvia (No. 71-43/87). Participation was voluntary, and informed consent was obtained from all individual participants included in this study.

3. Results

To determine the reliability of the AI competence scale, Cronbach's alpha value was calculated. The analysis of the Likert scale's reliability indicates that the internal consistency is outstanding ($\alpha = .925$) (Taber, 2018).

3.1. Competence Variation by AI function

Public sector employees self-assessed their AI competencies well below the scale's mean value (Table 2).

Only three AI sub-competencies have a mean value above 2: obtaining factual information in the form of Q&A (mean = 2.23, SD = 1.3), text translation or summarisation (mean = 2.16, SD = 1.35), and new text creation (mean = 2.13, SD = 1.30). This indicates that public sector employees most actively use AI as a source of information, similar to how they would use a search engine. The second-highest rated AI sub-competence, text translation or summarisation, has been around for some time, which may explain why its evaluation is slightly higher compared to other AI sub-competencies.

The mean values for the other seven AI sub-competencies are lower, all below 2. Additionally, the median for all seven of these AI sub-competencies is 1, indicating that at least half of the public sector employees have either never heard of these AI functions or have heard about them but lack specific knowledge. The three lowest-rated AI sub-competencies are audio creation (mean = 1.21, SD = .93), automation of routine work (mean = 1.22, SD = 1.07), and the prediction of sales figures, consumer behaviour, or quantitative data analysis for data-driven decision making (mean = 1.01, SD = 1.07).

The analysis of public employees' self-assessment distribution across competence levels suggests that many research participants have not tried most of the AI functions (Fig. 1).

Between 8 % and 32 % of research participants have not heard of certain AI functions. In particular, sales figures and consumer behaviour prediction, quantitative data analysis for data-driven decision making, and the automation of routine work are AI sub-competencies that at least 24 % of participants have never heard of. Additionally, many participants have heard about specific AI functions but only know of their existence without any deeper knowledge. This percentage ranges from 25 % to 52 %, depending on the AI sub-competence.

Proficiency in an AI sub-competence is considered to begin when self-assessment scores exceed 4, which indicates that respondents deliberately analyse their daily work and use AI tools to perform tasks more efficiently. There are three AI sub-competencies where this proficiency level is achieved more frequently: AI in the form of Q&A for obtaining factual information (51 %), new text creation (42 %), and text translation or summarisation (43 %). For the other seven measured AI sub-competencies, between 7 % and 19 % of participants have self-assessed their skills at level 4 or higher.

The results indicate that respondents tend to use AI functions with which they are already more familiar. The differences between self-assessments of AI sub-competencies are statistically significant, supporting the acceptance of hypothesis H1.

3.2. AI Training Effect

To understand whether the workplace provides necessary training for employees, the question "Is there AI training available in your workplace?" was posed to all research participants, with the following possible answers: 1 – no, 2 – rather no, 3 – rather yes, 4 – yes. The results indicate that most respondents do not receive the necessary training to improve their AI competence (Median = 2) (Table 3).

Additionally, AI training opportunities do not significantly differ among different age groups ($H(4) = 4.591$, $p = .332$; Appendix A) or education levels ($H(2) = 5.339$, $p = .069$; Appendix B). The results also indicate that employees who have received AI training have assessed their overall AI competence higher, and this difference is statistically significant (Table 4).

Respondents who answered positively to the question "Are AI training opportunities available in your workplace?" also self-assessed their AI competence at a higher level, and the difference is statistically significant. Therefore, hypothesis H2 should be accepted.

To understand whether AI sub-competencies are interconnected, the Spearman's rank correlation test was conducted. Results indicate that AI sub-competencies are interconnected (Table 5).

Across all possible AI sub-competence pairs, there is at least a moderate correlation ($rs > .3$) (Akoglu, 2018). Furthermore, 17 out of 45 possible pairs show a strong correlation ($rs > .6$). This suggests that

Table 1
Characteristics of study participants.

Age		**Type of populated point**	
18–29	99	Capital (Riga)	308
30–39	321	Cities (Ventspils, Liepāja, Jūrmala, Jelgava, Rēzekne, Daugavpils, Valmiera, Ogre, Jēkabpils)	326
40–49	431	Other towns	558
50–59	464	Countryside	365
60–74	236	**Education**	
75+	2	Basic education	2
Don't want to specify	4	Secondary education	44
Sex		Vocational education	87
Male	248	Higher education	1424
Female	1300		
Don't want to specify or other	9		

Table 2
Public sector employees' AI competence self-assessment.

AI sub-competence	Mean	Median	Standard deviation	Skewness	Kurtosis	Asymp. Sig.
1. In the form of Q&A for obtaining factual information	2.23	3	1.30	.01	−.65	<.001
2. New text creation	2.13	2	1.30	.37	−.44	
3. Text translation or summarisation	2.16	2	1.35	.37	−.44	
4. Speech or scanned document conversion to text	1.54	1	1.12	.78	.74	
5. Video creation	1.27	1	.95	.98	1.96	
6. Audio creation	1.21	1	.93	1.22	2.89	
7. Objects in images identification or images/videos marking and classification	1.33	1	1.11	1.05	1.33	
8. Motion or facial recognition	1.27	1	1.01	1.06	1.87	
9. Sales figures, consumer behaviour prediction or quantitative data analysis and data-based decision making	1.01	1	.97	1.34	2.98	
10. Routinised work automation	1.22	1	1.07	1.23	2.13	

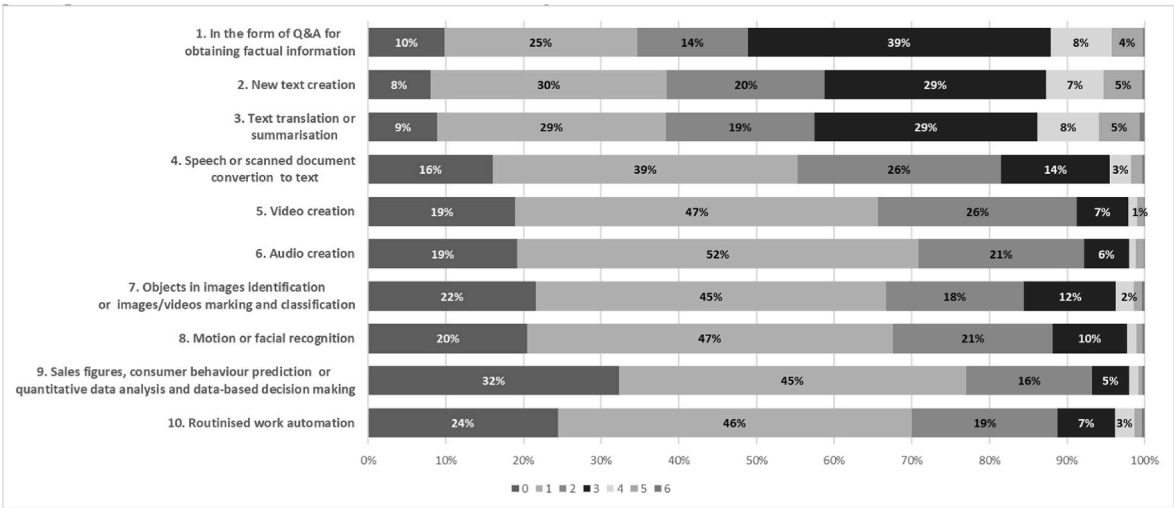

Fig. 1. Relative distribution of participants' AI sub-competence self-assessments.

Table 3
Is AI training available in your workplace?

Mean	Median	Standard deviation	Skewness	Kurtosis
2.11	2	1.08	.44	−1.16

Table 4
AI competence based on training opportunities.

Are AI training opportunities available in your workplace?	N	Mean	Median	Standard deviation	Asymp. Sig.
No	615	1.23	1.1	.80	<.001
Rather no	372	1.62	1.5	.85	
Rather yes	348	1.81	1.8	.86	
Yes	222	1.80	1.8	.85	

AI competencies are interconnected. However, since AI is a relatively new technology for the public and a broader introduction to it has only recently begun, this could also indicate that those who have started learning about AI tools are not limiting themselves to one functionality but are experimenting with several.

The network graph shows that 10 variables (nodes) form mutually significant connections, with all 45 non-zero edges formed, resulting in a sparsity of .000 (Fig. 2). Nodes in the graph are only connected if a correlation exists between them, and this covariance cannot be explained by any other variable in the network (Dimdiņš & Vanags, 2023; Lastovska & Lāma, 2024). Blue lines represent positive correlations, while red lines indicate negative correlations. The thickness of the line represents the strength of the correlation. It can be concluded that the strongest connection is between "Q & A for obtaining factual information" and "new text creation." These two AI sub-competencies are closely linked thematically, as created text could be enhanced with factual information acquired through Q&A interactions with an AI tool. Additionally, a strong link is observed between "sales figures, consumer behaviour prediction or quantitative data analysis" and "data-based decision-making and routinised work automation". Further research is required to uncover the reasons behind this connection. However, the findings indicate that improving one AI sub-competence is likely to have a positive impact on other AI sub-competencies, thereby highlighting the importance of training.

3.3. Trust Effect

To determine whether research participants trust AI tools, the question "Do you trust the information generated by AI?" was asked, with four possible choices: 1 – no, 2 – rather no, 3 – rather yes, 4 – yes (Table 6). The results indicate that more than half of the participants do not trust AI-generated information (Median = 2).

Trust in AI-generated information does not significantly differ among respondents with different education levels ($H(2) = 5.635$, $p = .060$; Appendix C). However, a statistically significant difference is observed across age groups ($H(4) = 27.476$, $p < .001$; Appendix D), with older respondents demonstrating lower levels of trust in AI-generated information compared to their younger counterparts.

The results also indicate that research participants who trust AI-generated information have assessed their overall AI competence

Table 5
Spearman's rank correlation between AI sub-competencies.

	2. New text creation	3. Text translation or summarisation	4. Speech or scanned document conversion to text	5. Video creation	6. Audio creation	7. Objects in images identification or images/videos marking and classification	8. Motion or facial recognition	9. Sales figures, consumer behaviour prediction or quantitative data analysis and data-based decision making	10. Routinised work automation
1. In the form of Q&A for obtaining factual information	.751**	.681**	.515**	.464**	.439**	.448**	.357**	.389**	.481**
2. New text creation		.692**	.525**	.483**	.465**	.458**	.341**	.409**	.487**
3. Text translation or summarisation			.633**	.538**	.481**	.485**	.429**	.441**	.526**
4. Speech or scanned document conversion to text				.707**	.601**	.600**	.553**	.541**	.573**
5. Video creation					.697**	.625**	.580**	.594**	.594**
6. Audio creation						.713**	.644**	.649**	.643**
7. Objects in images identification or images/videos marking and classification							.698**	.627**	.612**
8. Motion or facial recognition								.646**	.604**
9. Sales figures, consumer behaviour prediction or quantitative data analysis and data-based decision making									.780**

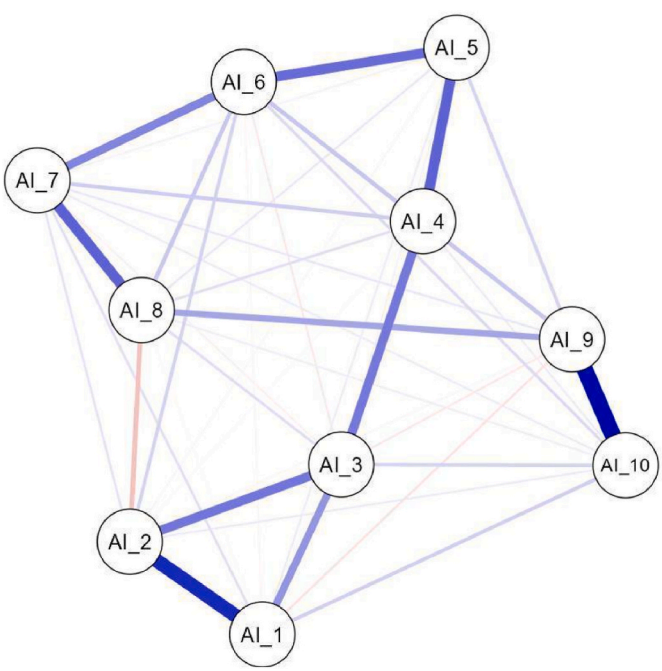

Fig. 2. Network analysis graph.

Table 6
Trust in information generated by AI.

Mean	Median	Standard deviation	Asymp. Sig.	Mean
2.38	2	.62	−.37	−.56

higher, and this difference is statistically significant (Table 7).

Respondents who indicated trust in AI-generated information reported higher self-assessed AI competence levels. Accordingly, hypothesis H3 is accepted.

The results also indicate that only 7 research participants fully trust AI. This is not necessarily negative, as AI can make errors. Future research on AI competence and trust in AI-generated results should be conducted.

4. Discussion

This study provides valuable insights into the current state of AI competence among public sector employees in Latvia, highlighting both functional disparities and the factors that shape self-assessed proficiency. Consistent with Hypothesis H1, the findings demonstrate a clear variation in AI competence across different functional domains. Employees reported greater familiarity and competence in basic AI tasks such as information retrieval, text creation and translation – tools that are widely accessible and often integrated into daily workflows. In contrast, more advanced applications, such as predictive analytics, image recognition and automation, were associated with lower competence levels. These results align with previous research suggesting that AI adoption typically begins with low-complexity tools that

Table 7
AI competence based on participant attitude.

Do you trust the information generated by AI?	N	Mean	Median	Standard deviation	Asymp. Sig.
No	111	1.02	1	.98	<.001
Rather no	752	1.39	2	1.07	
Rather yes	687	1.78	3	.96	
Yes	7	1.77	4	1.03	

enhance routine efficiencies and demands minimal technical specialisation (OECD, 2024a, 2024b; Wirtz et al., 2018). This trend is also consistent with innovation diffusion theory, which posits that simpler technologies tend to be adopted earlier than those requiring substantial organisational and cognitive investment.

Support was also found for Hypothesis H2, which identified a positive relationship between AI training and competence. Respondents who had access to training opportunities reported significantly higher self-assessed AI competence. This finding reinforces the critical role of facilitating conditions, a core dimension of the UTAUT framework (Venkatesh et al., 2003), and emphasises the importance of structured learning environments. The results mirror broader trends in digital transformation, where the lack of systematic training remains a key barrier to the effective use of technology in the public sector (McKinsey & Company, 2025). Given the rapid evolution of AI tools, continued investment in high-quality, job-relevant training will be essential to build the skills needed for effective implementation and sustained engagement.

In line with Hypothesis H3, the study also revealed a significant relationship between trust in AI-generated information and AI competence. Participants who expressed higher levels of trust in AI outputs tended to assess their AI competence more positively. This finding echoes the literature on human-AI interaction, which underscores trust as a foundational enabler of technology engagement (Glikson & Woolley, 2020; Siau & Wang, 2018). Importantly, the data also point to a generational divide: older employees reported lower levels of trust compared to their younger peers. This is consistent with prior research on digital confidence and adoption behaviour, which shows that age often moderates the willingness to experiment with and rely on emerging technologies (Eurofound, 2021; Schwab & Zahidi, 2020). These findings suggest that building trust is not only a technical or ethical issue but also a demographic and cultural one – necessitating differentiated strategies that address psychological barriers and promote inclusive digital participation.

Taken together, the study highlights the multifaceted nature of AI competence development in the public sector. Competence is shaped not only by functional familiarity but also by access to training, trust in technology and broader organisational culture. Compared to the private sector – where competitive pressures have driven faster AI integration and greater investments in workforce upskilling – the public sector faces additional challenges, including ethical scrutiny, resource constraints and slower innovation cycles (Chinn et al., 2020; OECD, 2024a, 2024b; European Commission: Directorate-General for Research and Innovation and Mazzucato M., 2018). Effective public sector strategies must therefore address both structural and attitudinal enablers of competence building.

Furthermore, network analysis results suggest that AI sub-competencies are interconnected, indicating that competence develops in a cumulative and reinforcing manner. For example, competence in basic text generation tools may serve as a gateway to mastering more complex AI applications such as automation or analytics. This supports staged or scaffolded learning models (Berryhill et al., 2019), where initial success builds confidence and lays the foundation for deeper, more diverse skill development. Such models can be particularly effective in public sector environments, where stepwise training can accommodate varied baseline skill levels and organisational capacities.

These findings offer clear policy implications. To enhance public sector AI competence, governments should expand access to AI training, tailor learning experiences to different roles and demographic groups, and foster organisational cultures that support digital innovation and ethical AI use. Strengthening trust-building mechanisms – through transparent AI design, explainability and consistent communication – will also be essential. Addressing these factors holistically will help ensure that AI is not only implemented but meaningfully and responsibly adopted in public service delivery.

In conclusion, AI competence in the public sector cannot be developed in isolation. It depends on an ecosystem of training, trust, technology access, and organisational readiness. Public institutions must embrace multi-dimensional strategies to cultivate a workforce that is not only capable of using AI but also confident, ethical and motivated to do so.

5. Conclusions

This study has provided important insights into the self-assessment of AI competence among public sector employees in Latvia. The findings show that while public sector employees are beginning to interact with AI tools, their overall competence remains low, particularly in more advanced AI functions. Only three AI sub-competencies – factual information retrieval, text translation and new text creation – scored close to the scale's mean value, demonstrating the limited scope of AI use within the public sector.

One key conclusion is that access to AI training significantly improves employees' competence levels. Those who received training rated their AI competence higher, underscoring the importance of structured training programmes in fostering AI adoption. Another critical finding is the low level of trust in AI-generated information, which may impede the broader adoption of AI tools. This distrust emphasises the need for organisations to promote transparency and ethical standards in AI deployment.

In conclusion, the study highlights the necessity of targeted AI training and trust-building initiatives to enhance AI competence and its application in the public sector. By investing in these areas, public sector organisations can fully leverage AI technologies to improve efficiency and decision-making processes. The results of this study provide a foundation for future research and policymaking aimed at addressing AI competence gaps and improving the integration of AI tools in public sector functions.

5.1. Limitations

While this study offers valuable insights into AI competence and trust among public sector employees in Latvia, several limitations should be acknowledged to contextualise the findings.

Firstly, the study relies on self-assessment as the primary method for measuring AI competence. Although self-reports are useful for capturing perceived proficiency, they are inherently subjective and may not accurately reflect actual skill levels. Participants may either overestimate or underestimate their capabilities, particularly regarding more complex AI functions that are not widely understood or used.

Secondly, the research adopts a cross-sectional design, capturing attitudes and competencies at a single point in time. This approach limits the ability to make causal inferences or to observe changes in competence and trust over time. Longitudinal studies would be better suited for tracking the evolution of AI skills and the effectiveness of interventions such as training programmes.

Thirdly, the study is limited to the Latvian public sector, which may affect the generalisability of results to other national contexts. Cultural, political and infrastructural differences could significantly influence how AI is adopted, perceived and integrated in public institutions elsewhere. Broader comparative studies across countries or sectors could provide more generalisable findings.

Fourthly, while the study examines AI competence across ten subdomains, the measurement was constrained to a single item per domain. Although based on Bloom's taxonomy, single-item measures may not fully capture the depth and complexity of each sub-competence. Future research could develop multi-item scales for each domain to increase construct validity and reliability.

Fifthly, although the survey reached over 1500 public employees, nonresponse bias cannot be entirely ruled out. It is possible that those more familiar with or interested in AI were more likely to participate, potentially skewing the data towards a more digitally literate sample

than the overall public sector population.

CRediT authorship contribution statement

Gatis Lāma: Writing – review & editing, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Agnese Lastovska:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Formal analysis, Conceptualization.

Data availability statement

The data that support the findings of this study are available on request from the corresponding author.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used AI for language revision (Chat GPT, Hugo.lv).

Funding statement

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Training opportunities by age group

Age group	N	Mean Rank	Kruskal-Wallis H	df	Asymp. Sig.
18–29	99	742.01	4.591	4	.332
30–39	321	800.02			
40–49	431	746.57			
50–59	464	794.88			
60–74	236	774.23			

Appendix B. Training opportunities by education

Education	N	Mean Rank	Kruskal-Wallis H	df	Asymp. Sig.
Secondary education	44	631.20	5.339	2	.069
Vocational education	87	775.26			
Higher education	1424	782.70			

Appendix C. Trust in AI-generated information by age group

Age group	N	Mean	Median	Mean Rank	Kruskal-Wallis H	df	Asymp. Sig.
18–29	99	2.51	3	851.03	27.476	4	<.001
30–39	321	2.49	3	850.26			
40–49	431	2.40	2	789.45			
50–59	464	2.31	2	728.42			
60–74	236	2.28	2	712.51			

Appendix D. Trust in AI-generated information by education

Education	N	Mean Rank	Kruskal-Wallis H	df	Asymp. Sig.
Secondary education	44	705.52	5.635	2	.060
Vocational education	87	694.80			
Higher education	1424	785.32			

References

Akoglu, H. (2018). User's guide to correlation coefficients. *Turkish Journal of Emergency Medicine, 18*, 91–93. https://doi.org/10.1016/j.tjem.2018.08.001. Available online:.

Alnamrouti, A., Rjoub, H., & Ozgit, H. (2022). Do strategic human resources and artificial intelligence help to make organisations more sustainable? Evidence from non-governmental organisations. *Sustainability, 14*(12). https://doi.org/10.3390/su14127327. Article 7327. Available online:.

Anderson, L. W., Krathwohl, D. R., Airasian, P. W., Cruikshank, K. A., Mayer, R. E., Pintrich, P. R., Raths, J., & Wittrock, M. C. (2001). *A taxonomy for learning, teaching, and assessing: A revision of Bloom's taxonomy of educational objectives*.

Arrighi, G. (2005). Globalization and inequality. *Social Analysis: The International Journal of Social and Cultural Practice, 49*(2), 3–16.

Auziņa, I., Darģis, R., Saulīte, B., Grūzītis, N., Grasmanis, M., Spektors, A., & Stepanovs, K. (2022). Development of a specialized Latvian speech corpus and pronunciation dictionary for the linguistic analysis and systematic transcription of visual diagnostic examinations. *Letonica, 2022*(47), 244–263. Available online: https://doi.org/LTNC.2022.0047.I.A.R.D.B.S.N.G.M.G.A.S.K.S.244.263.

Baki, N. U., Rasdi, R. M., Krauss, S. E., & Omar, M. K. (2023). Employee competencies in the age of artificial intelligence: A systematic review from southeast Asia. *International Journal of Academic Research in Economics and Management Sciences, 12* (1), 34–53. https://doi.org/10.6007/IJAREMS/v12-i1/15891. Available online:.

Becker, C., & Fischer, M. (2024). Factors of trust building in conversational AI systems: A literature review. In H. Degen, & S. Ntoa (Eds.), *Lecture notes in computer science( ): 14735. Artificial intelligence in HCI*. Cham: Springer. https://doi.org/10.1007/978-3-031-60611-3_3. HCII 2024.

Berryhill, J., Kok, H., & Clogher, R. (2019). Hello, world: Artificial intelligence and its use in the public sector. *OECD working papers on public governance*. https://doi.org/10.1787/726fd39d-en. Available online:.

Boginskis, V., Zadoroznijs, S., Cernavska, I., Beikmane, D., & Sauka, J. (2023). Artificial intelligence effectivity in fracture detection. *Medicni Perspektivi, 28*(3), 68–78. https://doi.org/2307-0404.2023.3.288965.

Borins, S. (2001). Encouraging innovation in the public sector. *Journal of Intellectual Capital, 2*(3), 310–319. https://doi.org/10.1108/14691930110400128. Available online:.

Bormane, S., & Blaus, E. (2024). Artificial intelligence in the context of digital marketing communication. *Frontiers in Communication, 9*. https://doi.org/10.3389/fcomm.2024.1411226. Article 1411226. Available online:.

Bostan, I. (2025). Consolidating the role of AI in the economy and society: Combating the deepfake phenomenon through strategic and normative approaches—The case of Romania in the EU context. *Economies, 13*(5), 129. Available online: https://doi.org/economies13050129.

Calvino, F., & Fontanelli, L. (2023). A portrait of AI adopters across countries: Firm characteristics, assets' complementarities and productivity. *OECD science, technology and industry working papers, no. 2023/02*. Paris: OECD Publishing. https://doi.org/10.1787/0fb79bb9-en. Available online:.

Chinn, D., Kaplan, J., Lambert, T., & Schwaiger, C. (2020). *AI adoption advances, but foundational barriers remain*. McKinsey & Company. Available online: https://mckinsey.com.

Chiu, T. K. F., Ahmad, Z., Ismailov, M., & Sanusi, I. T. (2024). What are artificial intelligence literacy and competency? A comprehensive framework to support them. *Computers and Education Open, 6*. https://doi.org/10.1016/j.caeo.2024.100171. Article 100171. Available online:.

Cramarenco, R. E., Burcă-Voicu, M. I., & Dabija, D. C. (2023). The impact of artificial intelligence (AI) on employees' skills and well-being in global labor markets: A systematic review. *Oeconomia Copernicana, 14*(3), 731–767. https://doi.org/10.24136/oc.2023.022. Available online:.

Davidescu, A. A., Apostu, S.-A., Paul, A., & Casuneanu, I. (2020). Work flexibility, job satisfaction, and job performance among Romanian employees—Implications for sustainable human resource management. *Sustainability, 12*(15). https://doi.org/10.3390/su12156086. Article 6086. Available online:.

De Vries, H., Bekkers, V., & Tummers, L. (2016). Innovation in the public sector: A systematic review and future research agenda. *Public Administration, 94*(1), 146–166. https://doi.org/10.2139/ssrn.2638618. Available online:.

Dimdiņš, G., & Vanags, E. (2023). Network analysis of social measures, culture dimensions, and COVID-19 related behavioural choices. In *Proceedings of the 81st international scientific conference of the university of Latvia* (pp. 8–18). https://doi.org/10.22364/htqe.2023.01. Available online:.

Dosi, G., Virgillito, M. E., & Zeli, A. (2021). Skills, productive structure and economic performance: The long-run impact of learning in economic development. *Structural Change and Economic Dynamics, 57*, 223–238. https://doi.org/10.1016/j.strueco.2021.03.005. Available online:.

Durán, J. M., & Pozzi, G. (2025). Trust and trustworthiness in AI. *Philosophy and Technology, 38*(1). https://doi.org/10.1007/s13347-025-00843-2. Article 16. Available online:.

Echeberria, A. L. (2022). The impact of AI on business, economics and innovation. In *Artificial intelligence for business: Innovation, tools and practices* (pp. 67–96). Springer. https://doi.org/10.1007/978-3-030-88241-9_3. Available online:.

Eke, C. I., & Shuib, L. (2024). The role of explainability and transparency in fostering trust in AI healthcare systems: A systematic literature review, open issues and potential solutions. *Neural computing and applications*. Advance online publication. https://doi.org/10.1007/s00521-024-10868-x. Available online:.

Ersoy, E. (2024). EU artificial intelligence regulation. *Computational Intelligence and Blockchain in Complex Systems*, 113–119. Available online: https://doi.org/B978-0-443-13268-1.00001-7.

Eurofound. (2021). *COVID-19: Implications for employment and working life. COVID-19 series*. Luxembourg: Publication Office of the European Union.

European Commission. (2019). *Ethics Guidelines for Trustworthy AI. High-Level Expert Group on Artificial Intelligence*. Available online: https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai.

European Commission: Directorate-General for Research and Innovation and Mazzucato, M.. (2018). *Mission-oriented research & innovation in the european union – A problem-solving approach to fuel innovation-led growth*. Publications Office. Available online: https://data.europa.eu/doi/10.2777/360325.

Glikson, E., & Woolley, A. (2020). Human trust in artificial intelligence: Review of empirical research. *The Academy of Management Annals, 14*(2). https://doi.org/10.5465/annals.2018.0057. Available online:.

Gonzalez, V., Amo, L., & Smith, S. D. (2025). Building trust in AI: Exploring the impact of AI competence framing. In *Proceedings of the annual Hawaii international conference on system sciences* (pp. 526–535). Available online: https://www.scopus.com/inward/record.uri?eid=2-s2.0-105005137988.

Govindan. (2013). Non-experimental research amenable to nursing contexts. *Asian J. Nur. Edu. & Research, 3*(1), 25–28. Available online: https://ajner.com/AbstractView.aspx?PID=2013-3-1-7.

Grant, D. L., Moreland, J. C., & Lauriat, P. M. (1979). Competency profiles in professional education: A means for adapting curricula to changing market demands. *The Journal of Education for Business, 54*(3), 120–123.

Gunaratne, H., & Pappel, I. (2020). Enhancement of the e-invoicing systems by increasing the efficiency of workflows via disruptive technologies. *Communications in Computer and Information Science, 1349*, 60–74. Available online: https://doi.org/978-3-030-67238-6_5.

Hartle, F. (1995). *How to redefine your competencies to maximize employee potential*. Financial Times Management.

Kaplan, A., & Haenlein, M. (2019). Siri, siri, in my hand: Who's the fairest in the land? On the interpretations, illustrations, and implications of artificial intelligence. *Business Horizons, 62*(1), 15–25. https://doi.org/10.1016/j.bushor.2018.08.004. Available online :.

Krop, P., Koch, M. J., Carolus, A., Latoschik, M. E., & Wienrich, C. (2024). The effects of expertise, humanness, and congruence on perceived trust, warmth, competence and intention to use embodied AI. *Conference on human factors in computing systems – Proceedings* https://doi.org/10.1145/3613905.3650749. Article 316. Available online:.

Labanava, A., Dreyling, R. M., III, Mortati, M., Liiv, I., & Pappel, I. (2022). Capacity building in government: Towards developing a standard for a functional specialist in AI for public services. *Communications in Computer and Information Science, 1688*, 503–516. https://doi.org/10.1007/978-981-19-8069-5_34. Springer. Available online:.

Lastovska, A., & Lāma, G. (2024). Fostering initiative and action-oriented mindsets for a civically engaged society. *EDULEARN24 Proceedings*, 4374–4381. https://doi.org/10.21125/edulearn.2024.1093. Available online:.

Leme, D., Alves, E., Lemos, V., & Fattori, A. (2020). Network analysis: A multivariate statistical approach for health science research. *Geriatrics, Gerontology and Aging, 14*, 43–51. https://doi.org/10.5327/Z2447-212320201900073. Available online:.

Li, Y., Wu, B., Huang, Y., Liu, J., Wu, J., & Luan, S. (2025). Warmth, competence, and the determinants of trust in artificial intelligence: A cross-sectional survey from China. *International Journal of Human-Computer Interaction, 41*(8), 5024–5038. https://doi.org/10.1080/10447318.2024.2356909. Available online:.

Limante, A., & Sukyte, M. (2025). Comparative insights and future directions of AI in the courts of the Baltic states. *International Journal of Law and Information Technology, 33*, Article eaaf002. https://doi.org/ijlit/eaaf002.

Livanis, E., Matsatsinis, N. F., & Kitsios, F. C. (2023). Artificial intelligence and operational research towards finance management: A research agenda. In *Springer proceedings in business and economics* (pp. 179–186). Springer. https://doi.org/10.1007/978-3-031-24294-6_19. Available online:.

Loucks, J., Hupfer, S., Jarvis, D., & Murphy, T. (2019). *Future in the balance? How countries are pursuing an AI advantage. Deloitte center for technology, media and telecommunications*. Deloitte Insights. Available online: https://www2.deloitte.com/content/dam/insights/us/articles/5189_Global-AI-survey/DI_Global-AI-survey.pdf.

McClelland, D. C. (1993). Testing for competence rather than for intelligence. *American Psychologist, 28*(1), 1–14. https://doi.org/10.1037/h0034092. Available online:.

McKinsey & Company. (2025). Superagency in the workplace: Empowering people to unlock AI's full potential. Available online: https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/superagency-in-the-workplace-empowering-people-to-unlock-ais-full-potential-at-work.

Mehrotra, S., Jorge, C. C., Jonker, C. M., & Tielman, M. L. (2024). Integrity-based explanations for fostering appropriate trust in AI agents. *ACM Transactions on Interactive Intelligent Systems, 14*(1). https://doi.org/10.1145/3610578. Article 4. Available online:.

Nag, A., Bhatia, A., Sharma, A., Kumar, V., & Sharma, V. (2023). Role of artificial intelligence in business dynamics: Opportunities and challenges. In *Leveraging AI and emotional intelligence in contemporary business organizations* (pp. 276–286). IGI Global. https://doi.org/10.4018/979-8-3693-1902-4.ch016. Available online:.

Nillos, B. E. (2016). Performance expectancy, effort expectancy, social influence and facilitating conditions are factors that influence rural health workers in the use of wireless access for health and perception of behavior of their pregnant patients. *JPAIR Multidisciplinary Research., 24*(1), 16–31. https://doi.org/10.7719/jpair.v24i1.376. Available online:.

Nübler, I. (2016). New technologies: A jobless future or golden age of job creation?. *Research department working paper no. 13)*. International Labour Organization. Available online https://www.ilo.org/wcmsp5/groups/public/—dgreports/—inst/documents/publication/wcms_544189.pdf.

OECD. (2024a). *National AI policies and strategies*. OECD. Available online: https://oecd.org.

OECD. (2024b). *Using AI in the workplace: Opportunities, risks and policy responses, OECD artificial intelligence papers, No. 11*. Paris: OECD Publishing. https://doi.org/10.1787/73d417f9-en. Available online:.

Olimid, A. P., Georgescu, C. M., & Olimid, D. A. (2024). Legal analysis of EU artificial intelligence act (2024): Insights from personal data governance and health policy. *Access to Justice in Eastern Europe, 7*(4), 120–142. Available online: https://doi.org/AJEE-18-7.4-a000103.

Ostertagová, E., Ostertag, O., & Kováč, J. (2014). Methodology and application of the kruskal-wallis test. *Applied Mechanics and Materials, 611*, 115–120. Available online: https://doi.org/10.4028/www.scientific.net/amm.611.115.

Polyviou, A., & Zamani, E. D. (2023). Are we nearly there yet? A desires & realities framework for Europe's AI strategy. *Information Systems Frontiers, 25*(1), 143–159. https://doi.org/10.1007/s10796-022-10285-2. Available online:.

Salman, M., Saleem, M., & Ganie, S. (2020). The concept of competence: A thematic review and discussion. *European Journal of Learning and Development, 44*(6), 717–742. https://doi.org/10.1108/EJTD-10-2019-0171. Available online:.

Schneider, K. (2019). The development of competency in ancient Roman law. *Legal History Journal, 34*(1), 45–61.

Schwab, K., & Zahidi, S. (2020). *The future of jobs report*. World Economic Forum. Available online: https://www3.weforum.org/docs/WEF_Future_of_Jobs_2020.pdf.

Shafi, K. (2017). A machine competence based analytical model to study trust calibration in supervised autonomous systems. In *2017 9th international conference on advanced computational intelligence (ICACI)* (pp. 245–252). https://doi.org/10.1109/ICACI.2017.7974516. Available online:.

Siau, K., & Wang, W. (2018). Building trust in artificial intelligence, machine learning, and robotics. *Cutter Business Technology Journal, 31*(2), 47–53.

Sinkus, T., & Ozola, I. (2024). Engineering student perceptions of AI technology implementation in ESP. *Engineering for Rural Development, 23*, 381–389. Available online: https://doi.org/ERDev.2024.23.TF073.

Slišāne, A., Lāma, G., & Bernande, M. (2021). Knowledge valorisation in doctoral studies in Latvia: Entrepreneurship and the development of research competencies in the study process. *Acta Paedagogica Vilnensia, 47*, 193–210. https://doi.org/10.15388/ActPaed.2021.47.13. Available online:.

Slišāne, A., Lāma, G., & Rubene, Z. (2022). How is entrepreneurship as generic and professional competences diverse? Some reflections on the evaluations of university students' generic competences (students of education and bioeconomics). *Frontiers in Education, 7*. https://doi.org/10.3389/feduc.2022.909968. Article 909968. Available online:.

Szwed, A. (2022). The use of artificial intelligence in migration-related procedures in the european union – Opportunities and threats. *Procedia Computer Science, 207*, 3639–3645. Available online: https://doi.org/j.procs.2022.09.424.

Taber, K. S. (2018). The use of Cronbach's alpha when developing and reporting research instruments in science education. *Research in Science Education, 48*, 1273–1296. https://doi.org/10.1007/s11165-016-9602-2. Available online:.

Ting, D. S. W., Liu, Y., Burlina, P., Xu, X., Bressler, N. M., & Wong, T. Y. (2019). AI for medical imaging goes deep. *Nature Medicine, 25*(1), 31–35. https://doi.org/10.1038/s41591-018-0029-3. Available online:.

Van der Klink, M. R., & Boon, J. (2002). The investigation of competencies within professional domains. *Human Resource Development International, 5*(4), 411–424. https://doi.org/10.1080/13678860110059384. Available online:.

Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly, 27*(3), 425–478. https://doi.org/10.2307/30036540. Available online:.

Venkatesh, V., Thong, J. Y. L., & Xu, X. (2012). Consumer acceptance and use of information technology: Extending the unified theory of acceptance and use of technology. *MIS Quarterly, 36*(1), 157–178. https://doi.org/10.2307/41410412. Available online.

Waytz, A., Cacioppo, J. T., & Epley, N. (2010). Who sees human? The stability and importance of anthropomorphizing the self. *Perspectives on Psychological Science, 5*(3), 219–232. https://doi.org/10.1177/1745691610369336. Available online:.

Wilcox, Y. (2012). *Competence through the ages: Tracing its origins and evolution*. Doctoral Dissertation. Cambridge: Harvard University.

Wirtz, B. W., Weyerer, J. C., & Geyer, C. (2018). Artificial intelligence and the public sector—Applications and challenges. *International Journal of Public Administration, 42*(7), 596–615. https://doi.org/10.1080/01900692.2018.1498103. Available online:.

Voronov, V. V., Menshikov, V. V., & Ruza, O. P. (2023). Artificial intelligence: A catalyst for entrepreneurship education in the baltics. *Baltic Region, 15*(3), 45–65. vailable online: https://doi.org/2079-8555-2023-3-3.

Vitello, S., Greatorex, J., & Shaw, S. (2021). *What is competence? A shared interpretation of competence to support teaching, learning and assessment*. Cambridge University press & assessment.